# Arjun P.

AI Engineer · AI agents, developer tools and AI products

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## SUMMARY

AI engineer with **5 years** of experience, the last two on production AI. At **Qilin Labs**, I built a multi-tenant **AI voice-agent platform** as the sole engineer: a streaming STT → LLM → TTS pipeline at **1.3s p50**, RAG, FastAPI services and evals. On my own, I build AI agents and their tools in Rust: a web browser written from scratch that agents drive over MCP, an open-source bot platform, and an LLM code reviewer measured against CodeRabbit.

## EXPERIENCE

### Senior Frontend Engineer — Aug 2024 – Present

Qilin Labs · Remote

- Built a multi-tenant **AI voice-agent platform** as the sole engineer: LiveKit SIP telephony, streaming STT → LLM → TTS, RAG on Qdrant, FastAPI services and a Next.js 14 ops console (**31 routes, 107 components**).
- Cut round-trip latency to **1.3s p50** and STT latency from **235 to 118ms** with regional routing. Fixed the telephony audio mixer, so dropped agent audio fell from **4.37% to 0%**.
- Built call monitoring, agent configuration and quality evals, covered by **2,585 backend tests**.
- Led the CRA → Next.js 14 migration of a **20K-user** education platform: **Lighthouse 58 → 92**, page load **~10s → ~3s**, JS bundle **2.4MB → 720KB**.

### Software Engineer · Frontend Lead — Mar 2023 – Sep 2024

Hubspire Corp · Remote

- Refactored a **30k+ LOC** legacy React codebase: dependency cleanup, code splitting, tree shaking and CI optimization.
- Designed a React component system adopted across **3+ projects**, built TypeScript and GraphQL (Apollo) apps, and mentored 3 junior engineers to mid-level.

### React Developer — Feb 2022 – Mar 2023

Woxro Technology Solutions · Thrissur, Kerala

- Built React frontends for banking apps with real-time payment tracking and Razorpay, plus a shared form-validation and error-handling system.

### Web Developer — Jul 2021 – Feb 2022

Limenzy Technologies · Calicut, Kerala

- Delivered **10+** responsive interfaces from Figma and Adobe designs.

## PROJECTS

**bare** ↗ — A web browser written from scratch in Rust, for AI agents
Rust, MCP · Hand-written TLS, HTTP/2, HTML, CSS, layout and a JavaScript interpreter in **16 crates**. An MCP server lets AI agents test sites and read pages as text before a screenshot. **1,390+ commits**; passed **48 journeys** on public sites.

**reviewer** ↗ — AI code review measured against CodeRabbit
Rust, LLM evaluation · Scored on **52 pull requests** built from real bug fixes in 16 open-source projects: F1 **0.29 vs 0.31** for CodeRabbit, with **0.63 vs 1.18** false alarms per clean PR, at about $0.09 per review.

**deck** ↗ — An open-source, self-hosted Grok Bot
Rust, Axum, React, QEMU/KVM · Chat with AI bots that each work in a persistent VM desktop with their own accounts, permissions and budget. The one-bot runtime passed a strict end-to-end run in a real VM.

**EdTech assessment platform** — AI grading · client project
React, Expo, FastAPI, Supabase, Claude vision · Lead engineer across **three apps** on one backend. Teachers scan answer sheets on a phone, Claude vision grades them, and a teacher reviews or overrides every score.

**figma-connect** ↗ — Local Figma MCP server for AI agents
TypeScript, Playwright, SQLite/FTS5 · **35 MCP tools** for design reads, plus verify_node: pixel-diffs an agent's generated code against the live Figma node.

## SKILLS

**AI & Agents:** AI agents, MCP, LLM evaluation, RAG (Qdrant), Claude API (vision), Vercel AI SDK
**Voice AI:** LiveKit (SIP/WebRTC), Deepgram, Silero VAD, streaming STT → LLM → TTS
**Languages:** Rust, Python, TypeScript, Go, Kotlin
**Backend & Infra:** FastAPI, Axum, Node.js, PostgreSQL, SQLite, Redis, Supabase, Docker, QEMU/KVM
**Frontend & Mobile:** React, Next.js, TanStack Query, Tailwind CSS, shadcn/ui, React Native, Expo
**Testing & Tooling:** Vitest, Jest, Playwright, Cypress, GitHub Actions

## EDUCATION

**B.E. Mechanical Engineering** · Dhanalakshmi Srinivasan Engineering College (Anna University) · 2017 – 2021
Also: author of @astra-ui-lib/core, a React component library published on npm.